COURSE GUIDE & SYLLABUS · KG–GRADE 2 · AGES 5–7

Scratch Junior (K–2) Beginner: Little Coders

Arrows, repeats, talking and treasure — first coding adventures with picture blocks.

A first coding course for ages 5–7, designed for children who are still learning to read. Children move characters with picture arrow blocks, add numbers to blocks, start scripts with the green flag, spot patterns and use repeat loops, make characters talk, collect treasure, and find and fix bugs. Every lesson has a short animated video, a build-along for the ScratchJr app, picture-block puzzles that check themselves, a talk-and-move activity, a printable worksheet and a teacher plan. Projects: Trip to the Park, Dance Party, Treasure Hunt and a capstone Story Show.

What learners will be able to do

- Move a character with arrow blocks in the right order (sequence)
- Change the number on a block to move further
- Start a script with the green flag (events)
- Spot patterns and use a repeat block (loops)
- Make a character talk at the right moment
- Plan a path to collect every gem
- Find and fix a bug by watching and changing one block
- Make code shorter with numbers and repeats
- Plan, build and present a short story in ScratchJr

How every lesson works

Watch	A short animated explainer video with captions (and an optional read-aloud).
Learn	Learn cards with the real coding blocks, and a step-by-step build-along.
Try it (Code Lab)	Code in the built-in Scratch-style Code Lab with a live stage. Challenges are checked automatically and give a clear next step; hints unlock one at a time.
Puzzles & quick checks	Robot puzzles and questions that build understanding, with instant feedback.
Probe	Debugging and “new situation” tasks to show real understanding.
Projects	Each module ends with a creative project built in the Code Lab and reviewed by the teacher with a rubric.
Printables	A worksheet for every lesson, a project brief for every project, and a lesson plan with answer keys for the teacher.

Scope and sequence

#	Lesson	Big idea	Learners build
1	**Arrows Tell Me Where to Go** Use arrow blocks to move a character, and put blocks in order to reach a goal.	Sequence	Move the cat in ScratchJr
2	**Numbers on Blocks** Change the number on a block to move more than one square, and choose the fewest blocks.	Parameters (numbers on blocks)	Big moves in ScratchJr
3	**Project: Trip to the Park** Plan and build a short trip for a character using arrow blocks with numbers, then share it.	Project: sequences	Project: Trip to the Park
4	**Ready, Set, Green Flag!** Start a script with the green flag block, and explain that a script needs a start block to run.	Events (starting a program)	Start with the flag
5	**Repeat It!** 🔁 Use a repeat block to do the same moves many times and make code shorter.	Loops (repeat)	Climb the stairs with repeat
6	**Project: Dance Party** Make characters dance with repeating moves started by the green flag, and share the dance.	Project: loops and events	Project: Dance Party
7	**Talk Bubbles** 💬 Make a character say something with a speech-bubble block, at the right moment in a script.	Output (speech)	Make the cat talk
8	**Collect the Gems** 💎 Plan a path to visit every gem and use a pick-up block at each one.	Planning a path (algorithms)	Gem hunt
9	**Project: Treasure Hunt** Plan and build a treasure hunt where a character collects treasure, talks, and reaches a goal.	Project: sequences, loops and talking	Project: Treasure Hunt
10	**Bug Hunt** 🐞 Find and fix a mistake (a bug) in a script by watching what the character does.	Debugging	Squash the bug
11	**Short and Sweet Code** Use numbers and repeat blocks to solve puzzles with as few blocks as possible.	Efficient code	Shrink the code
12	**Capstone: My Story Show** Plan, build, test and present a short ScratchJr story with characters that move, talk and repeat, started by the green flag.	Capstone project	Project: My Story Show

Modules

Module 1 · Arrows and order
Picture arrows, numbers on blocks and first sequences. Project: Trip to the Park.

Big question: How do we tell a character where to go?

Module 2 · Start and repeat
The green flag and repeat loops. Project: Dance Party.

Big question: How do we start code and repeat a pattern?

Module 3 · Talk and collect
Speech bubbles and collecting gems. Project: Treasure Hunt.

Big question: How can characters talk and find things?

Module 4 · Fix it and show it
Debugging, short code and a capstone Story Show.

Big question: How do coders fix mistakes and share their work?

Assessment

Coding challenges are checked on the server by running the learner's program. Quick checks give instant feedback. Written answers and the four projects are reviewed by the teacher against 4-level rubrics (Beginning · Developing · Secure · Excellent).

What you need

- Tablet or computer with a modern browser
- Optional: the free ScratchJr app (iPad, Android tablet or Chromebook) for the projects
- Paper and crayons for maps and plans

Before you start

- None — made for children who are just starting to read

Safety and privacy

Children use only first names and never share photos of themselves or others outside the class. ScratchJr projects stay on the device. Young learners use the AI tutor only with an adult present and guardian consent.

What comes next

After this course: **Scratch Junior (K–2) Intermediate**, which goes deeper into clones, custom blocks, lists and bigger games.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.